Doob's Proof of MLE Strong Consistency: A Surprisal and Information Perspective

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2026

Abstract

We give a concise, self-contained exposition of Doob's proof of strong maximum likelihood consistency, formulated in terms of surprisal and relative information. This formulation clarifies the roles of identifiability, properness, and continuity, and highlights that consistency holds without uniform convergence of the log-likelihood. We also verify these conditions for the multivariate normal distribution, recovering the sample mean and biased sample covariance as the maximum likelihood estimators.

Keywords: maximum likelihood estimator, true parameter, relative information

1 Introduction

The consistency of the Fisher (1922) maximum likelihood estimator (MLE) is a basic result discovered more than a century ago. This method is typically treated as a special case of M-estimators, and derived via standard techniques dating back to Cramér (1946) and Wald (1949). A modern treatment is van der Vaart (1998). Because of its intuitive appeal and broad applicability, the method is a dominant tool of statistical inference.

In teaching this material, I found the Doob (1934) strong consistency proof of MLE convergence more motivated and accessible to students, particularly for students already familiar with information and entropy as part of a data science or machine learning course. The fact that the proof covers general parameter spaces and general state spaces with essentially no extra effort only adds to its appeal.

The contribution of this note is expository and pedagogical: We give a formulation of Doob's consistency argument in surprisal/information terminology, emphasizing that strong consistency holds without assuming uniform convergence, and we work out the multivariate normal example.

2 Surprisal and Information

A random variable X is sampled repeatedly, resulting in independent and identically distributed copies X_1, $X_2, \dots$ of X. We assume the density $p(X, \theta^*)$ of X is one of a parametric family $p(X, \theta)$ of positive functions, and we wish to recover θ^* from the samples X_1, X_2, $\dots$.

In this formulation, one may interpret the family $p(X, \theta)$ as a probe of the underlying expectation E_*. While $p(X, \theta)$ may be chosen arbitrarily, based on prior knowledge, we have no control over E_*. Given this, how do we use the samples X_1, $X_2, \dots$ to recover θ^*?

Define the *surprisal* Samson (1951), Tribus (1961)

$$S(X, \theta) = \log(1/p(X, \theta)).$$

This terminology is intuitive: low probability equals high surprisal. Assume $S(X, \theta)$ is integrable for all θ. Then it is natural to call a minimizer θ^* of the *expected surprisal* $E_*(S(X, \theta))$ a *true parameter*, and the corresponding density $p(X, \theta^*)$ a *true density*.

Equivalently, given θ^* and θ, the *relative information* is

$$I(\theta^*, \theta) = E_*(S(X, \theta)) - E_*(S(X, \theta^*)). \tag{1}$$

The relative information is the excess expected surprisal of θ over θ^*. Then θ^* is a true parameter iff $I(\theta^*, \theta) \geq 0$ for all θ.

While $p(X, \theta)$ have so far been arbitrary positive functions, we now specialize to the case where $p(X, \theta)$ are densities in the usual sense. We say θ^* is a *reference parameter* if

$$E_* \left(\frac{p(X, \theta)}{p(X, \theta^*)} \right) = 1 \qquad \text{for all } \theta.$$

Then the corresponding density $p(X, \theta^*)$ is a *reference density*.

Let θ^* be a reference parameter, and define E and E_θ by

$$E_*(f(X)) = E(p(X, \theta^*) f(X)) \qquad \text{and} \qquad E_\theta(f(X)) = E(p(X, \theta) f(X)).$$

Then the relative information takes the familiar form Donsker & Varadhan (1975), Dembo & Zeitouni (1998)

$$I(\theta^*, \theta) = E \left(p(X, \theta^*) \log \left(\frac{p(X, \theta^*)}{p(X, \theta)} \right) \right), \tag{2}$$

and $E_\theta(1) = 1$ for all θ. Since $j(p) = \log(1/p)$ is convex, by Jensen's inequality,

$$I(\theta^*, \theta) = E_* \left(j \left(\frac{p(X, \theta)}{p(X, \theta^*)} \right) \right) \geq j \left(E_* \left(\frac{p(X, \theta)}{p(X, \theta^*)} \right) \right) = j(E_\theta(1)) = j(1) = 0, \qquad \text{for all } \theta.$$

It follows that a reference density is a true density.

If θ is any other true parameter, then $I(\theta^*,\theta)=0$, hence, by strict convexity of $j(p)$, there is a scalar t such that $p(X,\theta)=t\cdot p(X,\theta^*)$. Since θ^* is a reference parameter, $t=1$, hence $p(X,\theta)=p(X,\theta^*)$: every true density equals the reference density.

It follows that there is at most one reference density; when there is a reference density, it is the unique true density. We say parameters θ, θ' *share a density* if $p(X,\theta)=p(X,\theta')$. When parameters do not share densities, there is at most one reference parameter; when there is a reference parameter, it is the unique true parameter.

The simplest example is as follows. A vector $\theta=(\theta_1,\theta_2,\dots,\theta_d)$ is a probability vector if its components are nonnegative and sum to one. Suppose $\theta^*=(\theta_1^*,\theta_2^*,\dots,\theta_d^*)$ is a probability vector and a random variable X satisfies $P_*(X=i)=\theta_i^*$, $i=1,2,\dots,d$. Then the mass function of X is $p(X,\theta^*)=\theta_X^*$. If θ is any other probability vector, the surprisal is $S(X,\theta)=\log(1/\theta_X)$, and

$$I(\theta^*,\theta)=\sum_{i=1}^{d}\theta_i^*\log(\theta_i^*/\theta_i). \tag{3}$$

In the form (2), $I(\theta^*,\theta)$ is more commonly known as Kullback-Leibler divergence van der Vaart (1998) or relative entropy Cover & Thomas (2006), Dembo & Zeitouni (1998) or both Bishop (2006), Murphy (2022), and appears as the rate function in Sanov's theorem Sanov (1957). We regard it as information, consistent with some authors Kullback (1959), Schervish (1995), since information and entropy are conceptual opposites: the Shannon (1948) entropy $-\sum_{i=1}^{d}p_i\log p_i$ is concave, while (3) is convex.

Turning back to the general setting, the *likelihood* and (empirical) *mean surprisal* are

$$L_n(\theta)=\prod_{k=1}^{n}p(X_k,\theta),\qquad S_n(\theta)=\frac{1}{n}\log(1/L_n(\theta))=\frac{1}{n}\sum_{k=1}^{n}S(X_k,\theta).$$

The mean surprisal is also called the negative log-likelihood.

By definition, θ^* minimizes $E_*(S(X,\theta))$. By the strong law of large numbers (LLN), $S_n(\theta)$ converges to $E_*(S(X,\theta))$ as $n\to\infty$, with probability one, for each θ. Hence it is natural to estimate θ^* by a minimizer of $S_n(\theta)$ or, equivalently, a maximizer of $L_n(\theta)$: For each $n\ge 1$, a *maximum likelihood estimator (MLE)*, if it exists, is a parameter $\hat\theta_n$, depending on the samples, maximizing $L_n(\theta)$ over all θ. In what follows, only E_*, $p(X,\theta)$, and true parameters enter; E, E_θ, and reference parameters play no role.

3 Consistency

In addition to integrability of $S(X,\theta)$, we make three assumptions. Our first assumption is

A (Identifiability)**.** *There is at most one true parameter.*

A follows if the expected surprisal is strictly convex. Also, as we saw above, A follows if there is a reference parameter and parameters do not share densities.

A sequence of parameters *subconverges* to θ if a subsequence converges to θ. If a sequence subconverges to θ, θ is a *limit point of the sequence.* A subset K of the parameter space is *compact* if every sequence in K has limit points in K.

A function $F(\theta)$ is *proper* if its sublevel sets $F(\theta)\le c$ are compact subsets of the parameter space, for every level c. Since a proper continuous function has a minimizer and $\hat\theta_n$ should minimize $S_n(\theta)$, this leads us to the second assumption

B (Properness)**.** *There is an $N\ge 1$, depending on the samples, and a proper $F(\theta)$, not depending on the samples, such that, with probability one,*

$$F(\theta)\le S_n(\theta),\qquad \textit{for all } \theta \textit{ and } n\ge N. \tag{4}$$

When the parameter space is compact, every continuous function $F(\theta)$ is proper, and the proof below shows assumption B is superfluous. We will need the pointwise LLN convergence $S_n(\theta) \to E_*(S(X,\theta))$ simultaneously for all parameters, in the sense

$$S_n(\theta) \to E_*(S(X,\theta)), \qquad \text{as } n \to \infty, \qquad \text{for all } \theta, \tag{5}$$

with probability one. We emphasize this is not uniform convergence, since the rate of convergence may vary across parameters θ. For this we need the third assumption

C (Continuity)**.** *The parameter space is separable and there is a continuous $\delta(\theta,\theta')$ and an integrable non-negative $g(X)$ satisfying*

$$|S(X,\theta) - S(X,\theta')| \le \delta(\theta,\theta')g(X) \qquad \text{and} \qquad \delta(\theta,\theta) = 0. \tag{6}$$

Under C, a standard separability argument establishes (5) with probability one. Under C, the expected surprisal $E_*(S(X,\theta))$ is continuous in θ. If moreover B holds, then by sending $n \to \infty$ in (4), $F(\theta) \le E_*(S(X,\theta))$ for all θ, hence the expected surprisal is proper. Under A, B, and C, there is a unique true parameter. Similarly, under B and C, the mean surprisal is continuous in θ and proper, hence an MLE $\hat{\theta}_n$ exists for $n \ge N$, with probability one.

These assumptions are particularly tractable for *exponential families* Cover & Thomas (2006), Murphy (2022), Schervish (1995), van der Vaart (1998), in which $S(X,\theta)$ is essentially a sum of products of the form $f(\theta)g(X)$. In particular, we verify these assumptions for the multivariate normal in §4.

In (5), there is no presumption of uniform convergence. This is a key feature of Doob's proof. The argument instead combines properness, the LLN, and the continuity estimate in (6) to control limit points of the MLE sequence.

Theorem (Consistency)**.** *Let X_1, X_2, ... be independent and identically distributed, and let $p(X,\theta)$ be a parametric family of positive functions. Under the above assumptions,*

1. *the expected surprisal is proper and there is a unique true parameter θ^*,*
2. *with probability one, an MLE sequence $\hat{\theta}_n$ exists for sufficiently large n,*
3. *with probability one, any MLE sequence converges to the true parameter, $\hat{\theta}_n \to \theta^*$.*

Proof. The first two items were derived above. For the third item, by the LLN,

$$\frac{1}{n}\sum_{k=1}^{n} g(X_k) \to E_*(g(X)), \qquad \text{as } n \to \infty, \tag{7}$$

with probability one. Fix samples X_1, X_2, ... for which (4), (5) and (7) hold. We show $\hat{\theta}_n \to \theta^*$ for these samples. By (4),

$$F(\hat{\theta}_n) \le S_n(\hat{\theta}_n) \le S_n(\theta^*), \qquad \text{for } n \ge N.$$

Since $S_n(\theta^*) \to E_*(S(X,\theta^*))$, the right side is bounded. Since $F(\theta)$ is proper, the sequence $\hat{\theta}_n$ lies in a compact set in the parameter space, and thus has limit points.

Suppose θ is a limit point of $\hat{\theta}_n$. Then $\hat{\theta}_n$ subconverges to θ. By (6),

$$S_n(\theta) - S_n(\theta^*) \le S_n(\theta) - S_n(\hat{\theta}_n) \le \delta(\theta,\hat{\theta}_n) \cdot \frac{1}{n}\sum_{k=1}^{n} g(X_k). \tag{8}$$

Since the left side of (7) is bounded as $n \to \infty$, and $\delta(\theta,\hat{\theta}_n)$ subconverges to zero, the right side of (8) subconverges to zero. By (1) and (5), the left side of (8) converges to $I(\theta^*,\theta)$, hence $I(\theta^*,\theta) \le 0$. But $I(\theta^*,\theta) \ge 0$ (θ^* is a minimizer of $E_*(S(X,\theta))$) hence $I(\theta^*,\theta) = 0$, showing $\theta = \theta^*$. Thus θ^* is the only limit point, hence $\hat{\theta}_n \to \theta^*$ as $n \to \infty$. ∎

4 Multivariate Normal

Let μ be in $\mathbf{R}^d$, Q a positive definite $d \times d$ matrix, $\theta = (\mu, Q)$, and let $v \cdot w$ be the dot product in $\mathbf{R}^d$. With X in $\mathbf{R}^d$, the *normal density with mean μ and covariance Q* is $p(X, \theta) = e^{-S(X,\theta)}$ where

$$S(X, \theta) = \frac{1}{2} \log \det(2\pi Q) + \frac{1}{2}(X - \mu) \cdot Q^{-1}(X - \mu).$$

This defines the parametric family.

A random variable X in $\mathbf{R}^d$ is *normally distributed with parameter $\theta^* = (\mu^*, Q^*)$* if its moment-generating function is

$$E_* \left(e^{v \cdot X}\right) = \exp\left(v \cdot \mu^* + \frac{1}{2} v \cdot Q^* v\right). \tag{9}$$

Suppose X is normally distributed with parameter $\theta^* = (\mu^*, Q^*)$ as in (9), and let X_1, X_2, X_3, ... be independent and identically distributed copies of X. We show θ^* is the true parameter, and we verify A, B, and C.

If Q, R are symmetric matrices, we say $Q \geq R$ if $v \cdot Qv \geq v \cdot Rv$ for all vectors v. Then $aI \leq Q \leq bI$ iff the eigenvalues λ of Q satisfy $a \leq \lambda \leq b$. In particular, Q is nonnegative definite, $Q \geq 0$, if the eigenvalues satisfy $\lambda \geq 0$, and positive definite, $Q > 0$, if they satisfy $\lambda > 0$.

The sum and product of the eigenvalues of a symmetric matrix Q are the trace and determinant of Q, $\text{trace}(Q)$ and $\det(Q)$. The determinant is multiplicative and the trace is monotone: When $Q \geq R$ and $Z \geq 0$,

$$\det(SZS) = \det S \cdot \det Z \cdot \det S, \qquad \text{trace}(QR) = \text{trace}(RQ), \qquad \text{trace}(QZ) \geq \text{trace}(RZ).$$

The norm squared $\|Q\|^2 = \text{trace}(Q^2)$ is the sum of the squares of the entries of Q. Since $\|Q\|^2$ is also the sum of the squares of the eigenvalues of Q, we have $Q \leq \|Q\| \, I$.

By reducing to eigenvalues, for Z symmetric, one has

$$1 + |\det(I + Z) - 1| \leq \exp\left(\sqrt{d}\, \|Z\|\right).$$

This implies the determinant $\det Q$ is continuous on positive definite matrices Q.

Let $\theta' = (\mu', Q')$, $g(X) = 1 + |X|^2$ and let $\delta(\theta, \theta')$ equal half of

$$|\log \det Q - \log \det Q'| + \left|Q^{-1}\mu - (Q')^{-1}\mu'\right| + \left\|Q^{-1} - (Q')^{-1}\right\| + \left|\mu \cdot Q^{-1}\mu - \mu' \cdot (Q')^{-1}\mu'\right|.$$

Since the determinant is continuous, $\delta(\theta, \theta')$ is continuous. By standard manipulations (6) follows, establishing C.

Proposition. *For $Q^* > 0$ and c scalar, let $\alpha = e^{-2c-1} \det(2\pi e Q^*)$ and*

$$F_*(Q, Q^*) = \frac{1}{2} \log \det(2\pi Q) + \frac{1}{2} \text{trace}(Q^{-1} Q^*), \qquad Q > 0.$$

Then $F_*(Q, Q^*)$ is minimized iff $Q = Q^*$, and $F_*(Q, Q^*) \leq c$ implies $\alpha Q^* \leq Q \leq Q^*/\alpha$.

Proof. Let $h(\lambda) = \log \lambda + 1/\lambda$, $\lambda > 0$. Then $h(\lambda)$ is minimized iff $\lambda = 1$ and $h(\lambda)$ is proper in the sense $h(\lambda) \leq \log(1/\alpha)$ implies $\alpha \leq \lambda \leq 1/\alpha$. Write $Q^* = S^2$ with $S > 0$, and define Z by $Q = SZS$. Suppose $F_*(Q, Q^*) \leq c$ and let λ_1, λ_2, ..., λ_d be the eigenvalues of Z. Then the proposition follows from

$$\log(1/\alpha) - 1 \geq 2F_*(Q, Q^*) - 2F_*(Q^*, Q^*) = \log \det Z + \text{trace}(Z^{-1} - I) = \sum_{i=1}^{d} (h(\lambda_i) - 1). \qquad \square$$

Using the moment generating function,

$$E_*(S(X, \theta)) = F_*(Q, Q^*) + \frac{1}{2}(\mu^* - \mu) \cdot Q^{-1}(\mu^* - \mu).$$

By the proposition, (μ^*, Q^*) is the unique minimizer of the expected surprisal, establishing A.

Now let $0 < \epsilon < 1/2$ and

$$F_\epsilon(\theta) = F_*(Q, (1-2\epsilon)Q^*) + \frac{1}{2}(1-\epsilon)(\mu^* - \mu) \cdot Q^{-1}(\mu^* - \mu), \qquad \theta = (\mu, Q).$$

By the proposition with $(1-2\epsilon)Q^*$ replacing Q^*, if $F_\epsilon(\theta) \le c$, then Q is confined to a compact subset of positive definite matrices and, subsequently, μ is confined to a compact subset of $\mathbf{R}^d$. Hence $F_\epsilon(\theta)$ is proper.

Given vectors v, w in $\mathbf{R}^d$, the *tensor product* $v \otimes w$ is the matrix whose ij-th entry is $v_i w_j$. Then $v \otimes v \ge 0$ and

$$R = v \otimes v \qquad \Longrightarrow \qquad \operatorname{trace}(QR) = v \cdot Qv.$$

If $u = \sqrt{\epsilon} v - w/\sqrt{\epsilon}$, then $u \otimes u \ge 0$, hence

$$v \otimes w + w \otimes v \le \epsilon v \otimes v + \frac{1}{\epsilon} w \otimes w. \tag{10}$$

With

$$Q_n(\mu) = \frac{1}{n}\sum_{k=1}^n (X_k - \mu) \otimes (X_k - \mu), \qquad \hat{\mu}_n = \frac{1}{n}\sum_{k=1}^n X_k, \qquad \hat{Q}_n = Q_n(\hat{\mu}_n),$$

let $\hat{\theta}_n = (\hat{\mu}_n, \hat{Q}_n)$. Then $\hat{\mu}_n$ and $\hat{Q}_n$ are the usual sample mean and the biased sample covariance.

By the LLN applied to their entries, with probability one, the matrix sequences

$$(\hat{\mu}_n - \mu^*) \otimes (\hat{\mu}_n - \mu^*) \to 0, \qquad Q^* - Q_n(\mu^*) \to 0, \qquad \text{as } n \to \infty.$$

If Q_n is any sequence of symmetric matrices with $Q_n \to 0$, then $\|Q_n\| \to 0$. Since $Q_n \le \|Q_n\| I$, given any $Q > 0$, there is $N \ge 1$ such that $Q_n \le Q$ for $n \ge N$.

Thus there is $N \ge 1$, depending on the samples, such that, with probability one,

$$(\hat{\mu}_n - \mu^*) \otimes (\hat{\mu}_n - \mu^*) \le \epsilon^2 Q^*, \qquad Q_n(\mu^*) \ge (1-\epsilon)Q^*, \qquad \text{for } n \ge N. \tag{11}$$

Fix samples X_1, X_2, ... for which (11) holds. We establish (4) for these samples and $F(\theta) = F_\epsilon(\theta)$.

Let $f_\epsilon(\mu) = (1-2\epsilon)Q^* + (1-\epsilon)(\mu - \mu^*) \otimes (\mu - \mu^*)$. Then $F_\epsilon(\theta) = F_*(Q, f_\epsilon(\mu))$. With $w = \hat{\mu}_n - \mu^*$ and $v = \mu - \mu^*$, using (10) and inserting $\epsilon = 0$ in $f_\epsilon(\mu)$,

$$v \otimes w + w \otimes v \le \epsilon(\mu^* - \mu) \otimes (\mu^* - \mu) + \frac{1}{\epsilon}(\hat{\mu}_n - \mu^*) \otimes (\hat{\mu}_n - \mu^*) \le \epsilon f_0(\mu)$$

for all μ and $n \ge N$. Writing $X_k - \mu = (X_k - \mu^*) - v$,

$$Q_n(\mu) = Q_n(\mu^*) - (v \otimes w + w \otimes v) + v \otimes v.$$

Combined with the above inequalities, this leads to

$$Q_n(\mu) \ge f_\epsilon(\mu), \qquad \text{for all } \mu \text{ and } n \ge N.$$

In particular, since $f_\epsilon(\mu) \ge (1-2\epsilon)Q^*$, we have shown $\hat{Q}_n > 0$ for $n \ge N$.

Since $S_n(\theta) = F_*(Q, Q_n(\mu))$ and $Q_n(\mu) \ge \hat{Q}_n$ by least squares, by the proposition,

$$S_n(\theta) = F_*(Q, Q_n(\mu)) \ge F_*(Q, \hat{Q}_n) \ge F_*(\hat{Q}_n, \hat{Q}_n) = S_n(\hat{\theta}_n),$$

for all θ and $n \ge N$. Thus $\hat{\theta}_n$ is the MLE for $n \ge N$, and

$$S_n(\theta) = F_*(Q, Q_n(\mu)) \ge F_*(Q, f_\epsilon(\mu)) = F_\epsilon(\theta), \qquad \text{for all } \theta \text{ and } n \ge N.$$

This establishes (4) and completes the derivation of B.

5 Disclosures

The author reports no funding was obtained for this work and there are no competing interests to declare. The author used a generative AI tool (Anthropic's *Claude Sonnet*) to assist with language editing and bibliographic searches. All AI-assisted content was reviewed, verified, and edited by the author, who takes full responsibility for the content, accuracy, and originality of this work.

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